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Chapter 3. Modeling System Workflows: Activity Diagrams

Use cases show *what* your system should do. Activity diagrams allow you to specify *how* your system will accomplish its goals. Activity diagrams show high-level actions chained together to represent a process occurring in your system. For example, you can use an activity diagram to model the steps involved with creating a blog account.

Activity diagrams are particularly good at modeling *business processes* . A business process is a set of coordinated tasks that achieve a business goal, such as shipping customers' orders. Some business process management (BPM) tools allow you to define business processes using activity diagrams, or a similar graphical notation, and then execute them. This allows you to define and execute, for example, a payment approval process where one of the steps invokes a credit card approval web service—using an easy graphical notation such as activity diagrams.

Activity diagrams are the only UML diagram in the process view of your system's model, as shown in **Figure 3-1**.

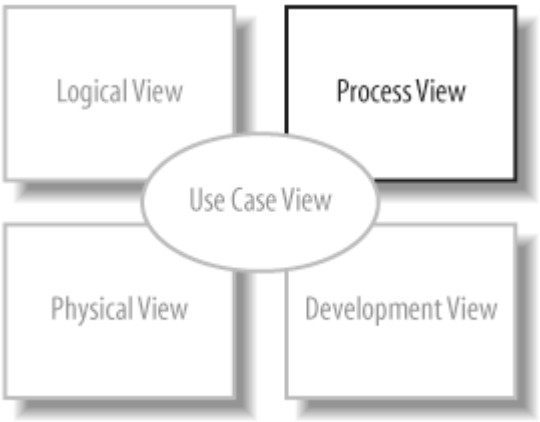


Figure 3-1. The Process View shows the high-level processes in your system—this is exactly what activity diagrams are good at doing

Activity diagrams are one of the most accessible UML diagrams since they use symbols similar to the widely-known flowchart notation; therefore, they are useful for describing processes to a broad audience. In fact, activity diagrams have their roots in flowcharts, as well as UML state diagrams, data flow diagrams, and Petri Nets.